Chandupa Marapana

+(94) 763953275 · chandupamarapana@gmail.com ·github · linkedin · Portfolio Website
Colombo, Srilanka.

SOFTWARE ENGINEERING UNDERGRADUATE

Software Engineering undergraduate with one year of industry experience at Cambio Software Engineering, contributing to a large-scale enterprise healthcare platform as part of the core framework team. Experienced in working with legacy Java codebases, collaborating with cross-functional teams, and delivering production-grade solutions in a clinical environment. Currently working on building software systems for manufacturing businesses using Spring Boot, React, and React Native. Expected graduation: November 2026.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, CSS
Frameworks: Spring Boot, Spring Security, React, React Native (Expo), Flask, Java Swing
Backend: REST API Design, JWT Authentication, Spring Boot Actuator, iText 7 (PDF)
Databases: MySQL, MongoDB, Supabase (PostgreSQL), Redis (basic)
DevOps & Cloud: Docker, Git, Railway, Vercel, Linux CLI, Agile / Scrum
ML & CV: YOLOv8, EfficientNet-B2, AlexNet, OpenCV
Testing: JUnit (unit testing), Postman, API testing

PROFESSIONAL EXPERIENCE

Cambio Software Engineering **Jun 2024 - Jun 2025**
Software Engineering Intern | Framework Team

• Contributed to a large-scale enterprise healthcare platform serving real clinical users, working within the core framework team whose components were shared across all product teams.
• Led a Java LTS version migration and Spring framework upgrade across a 50,000+ line legacy codebase, eliminating deprecated dependencies, resolving critical build failures, and restoring CI pipeline stability for a team of 8+ engineers.
• Collaborated with QA, product, and DevOps teams to resolve production bugs, refactor high-risk legacy modules, and improve code maintainability in a healthcare-critical system.
• Implemented i18n support for multi-language deployment, enabling the platform to serve international client rollout across multiple locales.
• Built foundational knowledge of Spring framework and Maven during the internship, which was independently applied to develop production Spring Boot systems post-internship.

KEY PROJECTS

Logmaster - Log delivery Management System · Spring Boot · MySQL · React · React Native · Docker
Ongoing Full-stack system commissioned by a Sri Lankan timber business to replace manual pen-and-paper log delivery billing. The backend implements versioned REST APIs (JWT auth, RBAC, Spring Boot Actuator), precise cubic-feet calculation using BigDecimal, and PDF bill generation via iText 7. Containerised with Docker; deployed to Railway (backend) and Vercel (web). (link)

Defect IQ - Real-Time Plywood Defect Detection System · Python · Flask · React · YOLOv8 · EfficientNet-B2
[Final Year Thesis] [AI / Computer Vision]

Dual-camera computer vision system that detects and classifies post-manufactured plywood surface defects in real time. Integrates a reasoning module to surface manufacturing root causes and preventive recommendations, enhancing quality control decision-making in industrial environments. Models: YOLOv8, EfficientNet-B2, AlexNet.(link)

Thread-Safe Exam Submission System· Java · Concurrent Programming
Built a concurrent exam submission system in Java using thread-safe coordination to handle multiple student submissions simultaneously. Implemented CountDownLatch and atomic integers to prevent race conditions, ensure data integrity, and support reliable processing under peak load with clean shutdown and monitoring. (link)

Thread-SafePatient Management System (A&E Simulation) · Java · Concurrent Programming
Multi-threaded A&E patient management simulation in Java with continuous patient arrivals, automated day/night shift rotation, and speciality-based consultant matching. Used thread-safe blocking queues and synchronised statistics to prevent data corruption under high-load conditions. (link)

GestureSway - Hand Gesture Snake Game· Python · React · MongoDB · Node.js
[5 member Group project]
Web application featuring a snake game controlled through real-time hand gesture recognition using machine learning. Designed to support patients with limited hand mobility. Built on the MERN stack with a Python ML backend. (link)

Wild Cafe Ella - Restaurant Website· HTML · CSS · JavaScript
[Freelance Group project]
Responsive promotional website for a jungle-style restaurant in Ella, Sri Lanka. Responsible for the menu, gallery, and contact pages. (link)

EDUCATION

Informatics Institute of Technology (IIT) Affiliated with University of Westminster **Sep 2022 - Nov 2026**
Bachelor of Engineering (Honours) in Software Engineering
Key modules: OOP, Concurrent Programming, Database Systems, Algorithms, Server-side Web Development, Cybersecurity, Operational Research.
Memberships: IEEE Club · IEEE Computer Society · Rotaract Club of IIT

Kegalu Vidyalaya **2019-2022**
GCE Advanced Level - Mathematics Stream

Royal International School **2018**
GCE Ordinary Level

ACHIEVEMENTS & EXTRACURRICULARS

•Tech Triathlon Designation Challenge -Hackathon organised by Rootcode Labs
•Member, IEEE Club & IEEE Computer Society - IIT
•Member, Rotaract Club of IIT
•Senior Prefect - Royal International School Kegalle

NON RELATED REFEREES

Sapna Kumarapathirage
Lecturer, Dept. of Computing
Informatics Institute of Technology,
Sri Lanka
(+94) 768209677· sapna.k@iit.ac.lk

Sakila Dissanayake
Associate Software Architect
Cambio Software Engineering
(+94) 777647504·
sakila.dissanayake@cambio.lk

I declare that the information provided in this CV is accurate and complete to the best of my knowledge.